

Python AI 101

Python + AI + 101

Class Overview

Class Overview: Python AI 101

Python

Python is the script
Interpreter for .py files

API

API is the fundamental
Application Programming
Interface for Programs to
Communicate

AI

AI allows API utilization
of AI resources, such as
an LLM

101

This is the beginning to
understand AND utilize
these fundamental
components



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Program

01

Python

Installing Python, IDE configuration and basic setup to write good Python code.

02

API

Selecting and using an Application Programming Interface, code-to-code communication.

03

AI

Using a Large Language Model, programmatically.

04

101

The most basic example of API+LLM to share structured input for unstructured output.

05

Prompt Tuning

Changing the personality of an unstructured prompt by adjusting natural language prompts.

06

Parameter Tuning

Changing the personality of an unstructured response by adjusting algorithmic inputs.

Python

Python is the programming/scripting language used in this presentation because it has a very low barrier-of-entry, shallow learning on-ramp, and massive exposure and use today.

Python is free to acquire, use, learn, and teach. With us here today, we use Python to embody the goal of this group to “MEET, LEARN, SHARE, and GIVE”.

There are numerous Python IDEs, Integrated Development Environments. As Python is not usually a compiled language, even a text-editor will do.

Distribution

www.python.org

Latest Version: 3.14.7 - released Aug 2026

New features: N-1 (3.13.x) - until 2029

Durable code: N-2 (3.12.x) - until 2028

Common IDEs:

- Jupyter Notebook
- VS Code
- PyCharm
- Kiro

API

Application Programming Interfaces include a wide range of ways for programs to communicate.

There have been in recent history a number of APIs.

We will concentrate on RESTful APIs.

Representational State Transfer (REST) is a software architecture that imposes conditions on how an API should work.

- Standard Format
- Statelessness

Methods:

- GET
- POST
- PUT
- DELETE

Source: What is REST? [[aws.amazon.com](https://aws.amazon.com/rest/)]

AI LLM

AI has generally made significant advancements over our lifetimes. The most recent major advancement in Artificial Intelligence has been the proliferation of Large Language Models (LLMs).

Large Language Models, using transformer architectures, have become remarkably adept at breaking down patterns in, and reproducing human language.

GPT stands for Generative Pre-trained Transformer - a type of LLM

Generative = generates text

Pre-trained = trained on copious amounts of text

Transformer = mathematically computes relationships between word parts (tokens)

A GPT continually “guesses” (computes) the next token.

We will utilize the capabilities of an LLM to generate responses from our unstructured prompts delivered with structured context.

101

The goal of this class is to bring about the basic concepts and knowledge to light.

This is not a how-to on vibe-coding an app.

The goal is to bring up the base level of understanding, to pave the way for new programmers to obtain valuable knowledge about the basic of using an LLM programmatically.

Structured Data vs Unstructured

Structured

XML	Tagged and hierarchical
YAML	Human-readable configuration
JSON	Key/value data pairs
TOON	Compact, token-efficient data

Unstructured

Words	Individual language elements
Instructions	Natural-language direction
Lists	Human-readable collections
Text	Free-form language
Markdown	Formatted plain text

(Natural-Language Context)

Natural Language Prompts

When we have a working use-case for AI LLM, we can extend the usefulness by using better models. A better model may be a simpler model, a larger model, a faster model. The use case and limitations work together.

Attempting different prompts allows important investigation in better, safer, more accurate, and more useful output.

Parameter Adjustment

Generation parameters influence how an LLM produces its response.

Temperature	variability of response
Top-P	consideration of possible next tokens
Max Tokens	limits maximum response length
Seed	helps produce repeatable results
Stop sequence	tells generation when to stop

(a small subset of common parameters)

Thank you



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